Top-down with left recursion

Np ⟶ det Adjh n

Adjh ⟶ Adjh adj

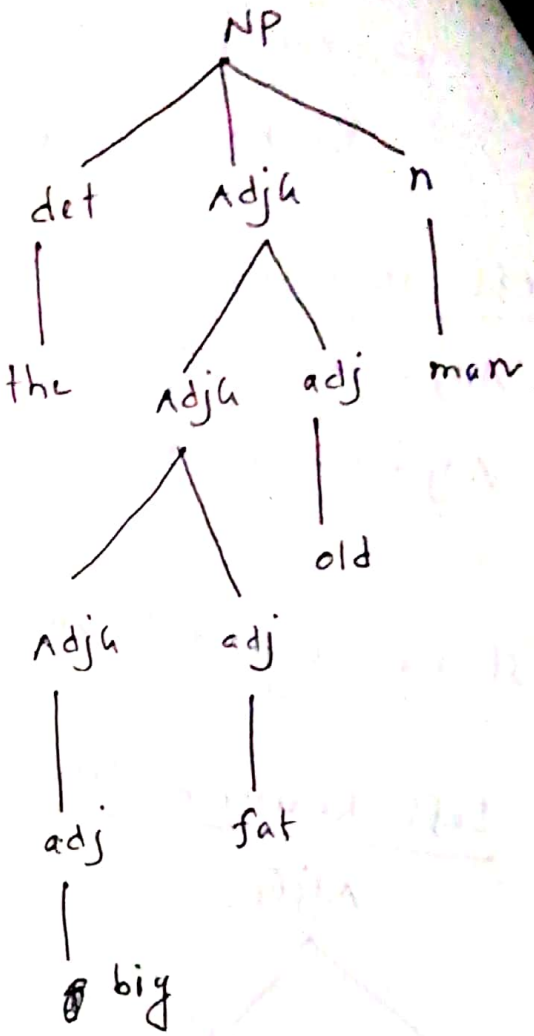
Adjh ⟶ adj

det ⟶ {an, the}

n ⟶ maw

adj ⟶ {big, fat, old}

string: the big fact rich old man

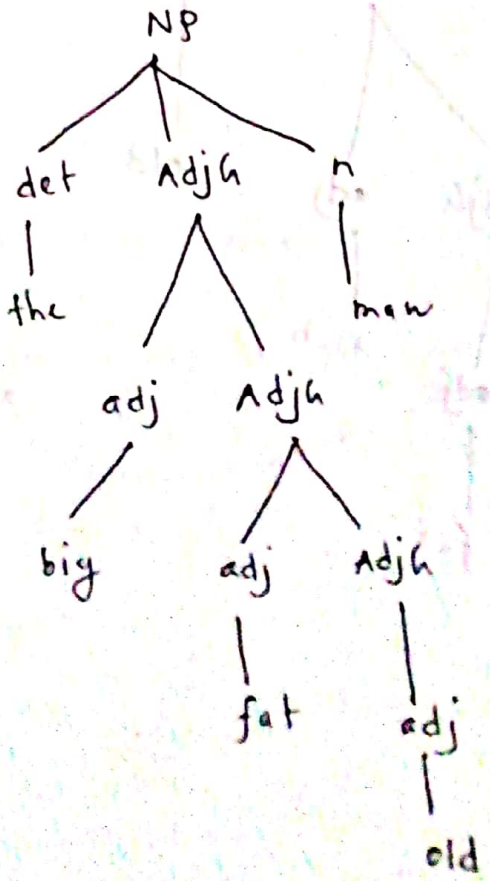


Top-down with right recursion

Np ⟶ det Adjh n

Adjh ⟶ adj Adjh

Adjh ⟶ adj



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Empty Rules

NP ⟶ det AdjG n

AdjG ⟶ adj AdjG

AdjG ⟶ ∈

Equivalent to

NP ⟶ det AdjG n | NP ⟶ det n

AdjG ⟶ adj AdjG | AdjG ⟶ adj

∴ NP ⟶ det AdjG n

NP ⟶ det n

AdjG ⟶ adj AdjG

AdjG ⟶ adj

Equivalent to

NP ⟶ det [AdjG] n → optional

AdjG ⟶ adj [AdjG] → optional

Top-down with empty rules

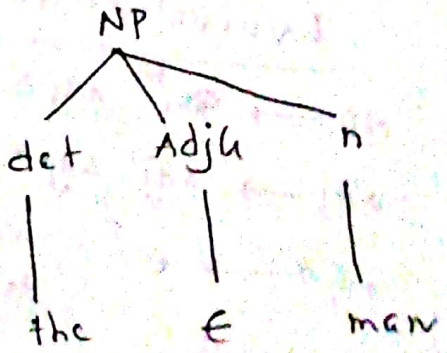
Grammar:

NP ⟶ det AdjG n

AdjG ⟶ adj AdjG

AdjG ⟶ ∈ , det → {the}
n → {man}

string: the man



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